

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2020

Bachelor in Computer Application
Course Title: Numerical Methods
Code No: CACS252
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 hrs

Candidates are required to answer all the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5=30]

2. Compute the root of equation $x^2 - 5x + 6 = 0$ in the vicinity of $x = 5$ using Newton-Raphson method.
3. Estimate the value of $\ln(2.5)$ using Newton-Gregory forward difference formula given the following data.

X	1.0	2.0	3.0	4.0
Ln(x)	0.0	0.6931	1.0986	1.3863

4. Write an algorithm to compute integration using Simpson's 1/3 rule. Evaluate integration of $\int_0^2 (3x^2 + 2x - 5) dx$ using Simpson's 1/3 rule. [2.5+2.5]
5. What is meant by ill-conditioned system? Find the Cholesky decomposition of the following matrix. [5]

$$\begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

6. Use the classical RK method to estimate $y(0.4)$ of the equation $\frac{dy}{dx} = x^2 + y^2$ with $y(0) = 0$ assume $h = 0.2$ [5]
7. Solve the Poisson equation $\nabla^2 f = 2x^2 y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$. [5]
8. Write a short note on (Any Two): [2.5+2.5]
 - a) Types of Errors

b) Convergent Methods

c) PDE

Group C

Attempt any TWO questions.

[2×10=20]

9. Write an algorithm and program to compute root of nonlinear equation using bisection method.

[4+6]

10. Given then data points:

I	0	1	2
X_i	9	16	25
f_i	3	6	2

Estimate the function value of f at x = 14 using cubic splines.

[10]

11. a) Solve the following system of equation by Gauss-Seidel method.

[5]

$$2x - 7y - 10z = -17$$

$$5x + y + 3z = 14$$

$$X + 10y + 9z = 7$$

b) Write an expansion of Taylor's Theorem. Solve the equation

$y'(x) = x^2 + y^2$, $y(0) = 1$ for $x = 0.5$ using Taylor method.

[2.5+2.5]